

The following outline is provided as an overview of and topical guide to space exploration.

Space exploration – use of astronomy and space technology to explore outer space. Physical exploration of space is conducted both by human spaceflights and by robotic spacecraft.

Essence of space exploration

Space exploration

Big science – Term used to describe a series of changes in science which occurred in industrial nations

Exploration – Investigating an unfamiliar area

High tech – Most advanced technology available

Space – Void between celestial bodies

Spaceflight – Flight into or through outer space

Branches of space exploration

Human spaceflight – Spaceflight with a crew or passengers

Uncrewed spaceflight – Autonomous space travel without human

History of space exploration

Remote sensing of Earth

Exploration of the Moon

Apollo program

Moon landings

Robotic exploration of the Moon

Exploration of Mercury

Exploration of Venus

Exploration of Mars

Mars landings

Mars rovers

Mars Rotorcrafts

Mars flyby

Exploration of Jupiter

Exploration of Saturn

Exploration of Uranus

Exploration of Neptune

History of human spaceflight

Project Mercury

Project Gemini

Apollo program

Space Shuttle program

Vostok program

Voskhod program

Soyuz program

Shenzhou program

List of human spaceflights

List of Space Shuttle missions

Spaceflight records

Emergence of market forces in spaceflight

Timeline of artificial satellites and space probes

Timeline of astronauts by nationality

Timeline of first orbital launches by country

Timeline of rocket and missile technology

Timeline of space exploration

Timeline of space travel by nationality

Timeline of spaceflight

Timeline of the Space Race

Timeline of Solar System exploration

Space agencies

List of government space agencies

Space agencies capable of human spaceflight (as of January 2024)

NASA (USA)

CNSA (China)

RFSA (Russia)

Space agencies with full launch capability

NASA (USA)

RFSA (Russia)

CNSA (China)

ISRO (India)

ESA (Europe)

JAXA (Japan)

ISA (Israel)

KARI (South Korea)

Active deep space missions and space stations

International Space Station

Space Weather Follow On-Lagrange 1 (NOAA)

Carruthers Geocorona Observatory (NASA)

Interstellar Mapping and Acceleration Probe (NASA)

Europa Clipper (NASA)

Tiangong space station (CNSA)

Chandrayaan-2 (ISRO)

Tianwen-2 (CNSA)

Hera (ESA)

Tiandu-1 and 2 (CNSA)

Queqiao relay satellite (CNSA)

Queqiao-2 relay satellite (CNSA)

ICUBE-Q (SUPARCO)

Chang'e 5 service module (CNSA)

DRO A/B (CAS)

Chang'e 4 (CNSA)

CAPSTONE (NASA)

Danuri (KARI)

Advanced Composition Explorer – NASA mission to observe solar wind

Deep Space Climate Observatory – NOAA observatory for space weather

Aditya-L1 (ISRO)

Mars Odyssey (NASA)

Mars Express – ESA satellite orbiting Mars

Mars Reconnaissance Orbiter (NASA)

Mars Science Laboratory – NASA rover to Mars

MAVEN – NASA satellite orbiting Mars

ExoMars Trace Gas Orbiter (ESA / Roscosmos) – Mars satellite

Tianwen-1 (CNSA)

Mars 2020 Perseverance rover – NASA rover to Mars

BepiColombo (ESA / JAXA)

STEREO – NASA mission to observe the Sun

Parker Solar Probe – NASA probe to Sun

Solar Orbiter – ESA probe to Sun

Hayabusa2# (JAXA) – sample return mission to asteroid Ryugu

OSIRIS-APEX (NASA) – probe to asteroid Apophis

Lucy – NASA probe to multiple Jupiter trojans

Psyche (NASA) – probe to asteroid Psyche

Juno – NASA satellite orbiting Jupiter

Jupiter Icy Moons Explorer – ESA probe to Jupiter and its moons

New Horizons – probe to Pluto

Voyager 1 and Voyager 2 (NASA) – probes to outer Solar System and interstellar space

Future of space exploration

Lunar (the Moon)

Future lunar missions

Colonization of the Moon

Lunar outpost (NASA)

Mercury

Colonization of Mercury

Venus

Exploration of Venus

Mars

Colonization of Mars

Human mission to Mars

Mars to Stay

Outer Solar System

Colonization of the outer Solar System

Colonization of Titan

Beyond the Solar System

Interstellar travel

Nuclear rocket

Fusion rocket

Solar sail

Einstein-Rosen bridge

Alcubierre drive

Intergalactic travel

General space exploration concepts

Astronaut – Spacecraft crew member

Non-rocket spacelaunch – Concepts for launch into space

Space and survival – Idea that spacefaring is necessary for long-term human survival

Space vehicle – Combination of launch vehicle and spacecraft

Launch vehicle – Rocket used to carry a spacecraft into space

Spacecraft – Vehicle or machine designed to fly in space

Spaceflight – Flight into or through outer space

Space research – Scientific studies carried out using scientific equipment in outer space

Space exploration scholars

Leaders in space exploration

Yuri Gagarin – first man in space

Neil Armstrong and Buzz Aldrin – first men to walk on the Moon

John Glenn – oldest man in orbit

See also

Outline of space science

Outline of aerospace

Timeline of Solar System exploration

Scientific research on the International Space Station

Lists

List of spacecraft

List of crewed spacecraft

List of Solar System probes

List of active Solar System probes

List of proposed Solar System probes

List of lunar probes

List of Mars landers

List of Mars orbiters

List of space telescopes

List of proposed space telescopes

List of cargo spacecraft

List of Falcon 9 first-stage boosters

List of heaviest spacecraft

List of spacecraft called Sputnik

List of spacecraft powered by non-rechargeable batteries

List of spacecraft with electric propulsion

List of spaceplanes

List of upper stages

List of spacecraft deployed from the International Space Station

Assembly of the International Space Station

Space Shuttle crews

List of Apollo astronauts

List of Apollo missions

List of Artemis missions

List of artificial objects on extra-terrestrial surfaces

List of astronauts by name

List of astronauts by selection

List of communication satellite companies

List of communications satellite firsts

List of Constellation missions

List of Cosmos satellites

List of crewed spacecraft

List of cumulative spacewalk records

List of Earth observation satellites

List of human spaceflight programs

List of human spaceflights

List of human spaceflights to the International Space Station

List of interplanetary voyages

List of ISS spacewalks

List of International Space Station expeditions

List of International Space Station visitors

List of landings on extraterrestrial bodies

List of launch vehicles

List of Mir expeditions

List of Mir spacewalks

List of NASA missions

List of objects at Lagrangian points

List of private spaceflight companies

List of probes by operational status

List of rockets

Lists of rocket launches

List of Ariane launches

List of Atlas launches

List of Black Brant launches

List of Falcon 9 and Falcon Heavy launches

List of Long March launches

List of Proton launches

List of R-7 launches

List of Scout launches

List of Space Launch System launches

List of Thor and Delta launches

List of Titan launches

List of V-2 test launches

List of Zenit launches

List of Russian human spaceflight missions

List of satellites in geosynchronous orbit

List of Solar System probes

List of Soviet human spaceflight missions

List of space agencies

List of Space Shuttle missions

List of space travelers by name

List of space travelers by nationality

List of spacecraft and crews that visited Mir

List of spacecraft manufacturers

List of spaceflight records

List of spaceports

List of spacewalks and moonwalks

List of the largest fixed satellite operators

Uncrewed spaceflights to the International Space Station

Lists of astronomical objects

Lists of telescopes

List of government space agencies

Lists of astronauts

Lists of space scientists

References

External links

NASA's website on human space travel

ESA: Building the ISS

Unofficial Shuttle Launch Manifest

ISS Assembly Animation Archived 2008-01-19 at the Wayback Machine

List of All Spacecraft Ever Launched, accessed 02/10/2019